

Dan-Sco: A Wearable Soft Sensor Prototype Designed for Live Dance-Music Performance

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Abstract

Dan-Sco is a wearable soft sensor prototype designed for live performance contexts. This paper focuses on the design and fabrication of two custom garments that integrate textile-based stretch sensors to capture full-body movement. Through an iterative prototyping process—including embroidery, knitting, conductive painting, and custom tailoring—we explored how different e-textile techniques and materials respond to dynamic motion. Each garment embeds soft sensors at targeted body zones to ensure both comfort and reliable signal capture during high-mobility activities. We detail the construction process, sensor material selection, garment integration strategies, and design decisions that balance wearability, durability, and signal responsiveness. This work offers practical insights into building expressive, movement-sensitive wearables using soft, textile-based materials.

CCS Concepts

• Human-centered computing → Interaction design; • Applied computing → Performing arts.

Keywords

wearable, soft sensors, interactive performance, embodied interaction

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1 Introduction

Contemporary dance highlights the expressive potential of the body, but real-time sound interaction through garments remains underexplored. Prior works such as *DanceBits*[2] and *Wearable Technology for Opera*[4] have used wearables for movement sensing or instructional feedback, yet few have embedded soft sensor systems directly into performance costumes for interactive sound generation.

Dan-Sco introduces a wearable soft sensor prototype that transforms garments into musical interfaces. By integrating conductive textiles into multidirectional stretch fabric, the system captures whole-body movement and maps it to musical parameters such as pitch, rhythm, and timbre in real time. This embodied interaction framework enables dancers to generate sound through physical gesture, turning the body itself into an expressive instrument.

This paper focuses on the design and fabrication of the sensor-embedded garments that make this interaction possible. We detail the prototyping process—from sensor material experimentation to garment construction and integration—aiming to provide practical insights for developing soft, body-responsive wearables.

2 Wearable Sensor Garments

The Dan-Sco system consists of two multidimensional stretch garments embedded with soft textile sensors. These sensors were crafted using conductive thread, conductive paint, and hand-sewn electrodes, placed in motion-intensive areas of the garments. Each

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garment was tailored for a specific dancer and calibrated to their individual movement vocabulary, ensuring that the system could responsively map full-body gestures to musical output in performance.

2.1 Wearable Sensor Experiments

Electronic textiles (e-textiles) incorporate electronic functionality directly into woven or knitted structures [7, 9]. Owing to their flexibility and conformability, e-textiles can be seamlessly embedded into garments and are widely applied in biosensing [1, 10], wireless communication [8, 11], and healthcare [10]. To identify fabrication techniques suitable for movement-responsive soft sensors, we extensively tested a range of textile processes—including weaving, knitting, embroidery, sewing, lacemaking, and screen printing—using various conductive materials. Each prototype underwent functional tests involving representative dance movements to evaluate signal quality, stability, and material compatibility. Based on performance, flexibility, and production feasibility, we selected sewing, freehand embroidery, and conductive painting on stretch fabric as our core fabrication methods. These approaches offered reliable sensor performance while retaining the material expressivity needed to capture nuanced, full-body dance gestures.

2.2 Garment Design

The design of the Dan-Sco sensor garments sought to balance playfulness, structural integrity, and kinetic responsiveness. To maximize freedom of movement and leverage the tension properties of spandex, we employed subtraction pattern cutting techniques to create two unconventional silhouettes. Each dress began as a basic pocket shape formed from two 1.5-square-yard panels of flat jersey, which were then sculpted into three-dimensional forms through subtraction and draping.

To enhance both the visual expressiveness and dynamic stretch behavior of the garments, whale boning was inserted along sections of the outer seams. These structural reinforcements supported sustained tension and preserved form during rapid motion, enabling the sensors to maintain contact and responsiveness. The result is a costume that functions as both a soft interface and a performative object, responsive to bodily input while visually amplifying movement on stage.

2.3 Collaborative Movement Design with Dancers

After completing the initial prototypes, we engaged two professional dancers to participate in an exploratory flexibility and interaction session. Dancers were invited to improvise freely while wearing the garments, allowing their embodied responses to shape the interaction possibilities. Three researchers observed and documented these sessions, focusing on areas of the garment that elicited the most movement and appeared most conducive to intuitive interaction. Based on these observations, we identified key zones for sensor placement. Following this, we introduced the underlying sensor interaction principles to the dancers and conducted a collaborative ideation session with the musician and garment designer. Together, we mapped potential movement types to sonic responses, forming the basis for a co-created movement-to-sound vocabulary.

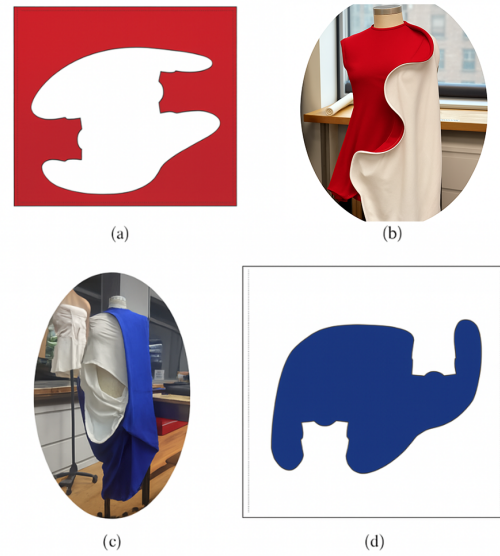


Figure 1: Design process for sensor-integrated garments in the Dan-sco project. (a) Subtraction pattern making for the red dress. (b) The draped red dress on a mannequin showing spatial structure. (c) Draping process of the blue dress on a mannequin showing spatial structure. (d) Subtraction pattern making for the blue dress.

3 Sensing and Sound Mapping

3.1 Soft Sensor on the Garment

Following a co-design process with dancers, we identified sensor locations and textile patterns on stretch fabric that best support expressive, movement-based interaction. As illustrated in Figure 3, the blue garment features embroidered sensors on the right shoulder and upper back, stitched with conductive thread. These placements were selected to capture rhythmic shoulder and back movements during performance. The embroidered leaf motif serves both aesthetic and functional purposes—its branching structure enhances sensitivity to directional movement and stretch variations. On the inner front panel, a hand-painted conductive pattern was applied to detect subtle changes in upper body movement. Additionally, seven conductive beads were embedded underneath the diagonal hem of the blue cape-part in the front. When the swinging hem contacts the conductive pattern on the adjacent white panel, it temporarily closes the circuit, producing trigger-like interactions akin to piano keys. This enables rhythmic sonic output generated through bodily impact.

As shown in Figure 4, the red garment features a continuous calligraphic pattern with conductive paint, extending from the front neckline to the hem. This vertically integrated design was developed to enhance the garment’s ability to sense the rhythm and dynamics of the dancer’s torso. The elongated conductive path captures variations in movement intensity and direction across a larger bodily surface, supporting more nuanced motion-to-sound mapping.

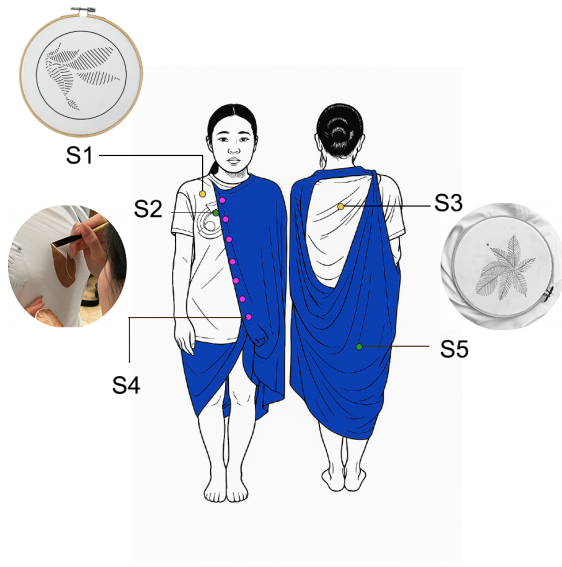


Figure 2: Sensor locations and design patterns on the blue garment. S1: Conductive thread leaf embroidery placed on the right shoulder for capturing shoulder motion. S2: Hand-painted conductive pattern at bust level detects upper torso movement. S3: Large-scale conductive leaf embroidery on the back, combining aesthetic motif with functional stretch sensing. S4: A row of seven conductive beads sewn along the inner edge of the cape forms a temporary closed circuit when contacting the chest pattern (S2), enabling rhythmic triggering akin to piano keys. S5: A linear conductive painting inside the blue panel on the back captures tension from dancer’s movements, converting movement into digital output and transform the signals into sound through real-time sonification.

3.2 System Design

Sensor data—including pressure, stretch, and bending—is collected by an Arduino Nano 33 IoT microcontroller paired with a Nano Wearable Shield[3, p. 151], embedded in each garment. These microcontrollers wirelessly transmit the sensor data via the MQTT protocol to a corresponding Arduino Nano 33 IoT module connected to a host computer. The receiving unit interfaces with Logic Pro via USB MIDI, translating motion-triggered sensor signals into real-time musical output. This modular, low-latency system architecture facilitates seamless and responsive movement-to-sound mapping, supporting improvisational performance dynamics.

3.3 Sound Mapping

The musician configured three channels in Logic Pro to mediate real-time interactions between the dancers’ embodied gestures and the evolving soundscape. One channel carried background music, while the other two were modulated by sensor signals embedded in the performers’ garments. In the blue one, the back sensor (S3) was mapped to BPM, enabling tempo modulation through full-body rhythm; the shoulder (S1) modulated pitch to support expressive

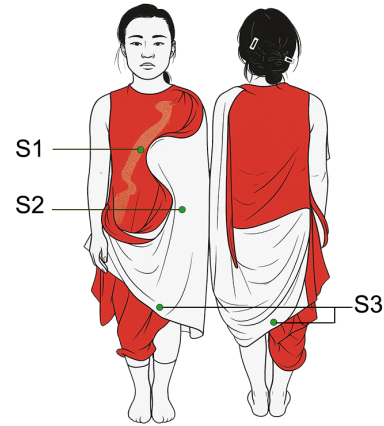


Figure 3: Sensor locations and design patterns on the red garment. S1: Painted sensor on the front panel of the garment, positioned to detect upper body movement. S2: Linear conductive paint sensor placed beneath the internal whalebone structure to capture torsional deformation. S3: Strip-shaped conductive sensor applied along the inner hemline, enabling detection of motion at the garment’s lower edge.

nuance; the cape (S4) triggered individual notes; S5 adjusted the dry/wet signal ratio to shape emotional texture; and the front pattern (S2) controlled volume. In the red garment, S1 (front) was mapped to pitch, S2 (whalebone) to timbral variation, and S3 to EQ. These mappings were not fixed but emerged through iterative rehearsal, where the musician’s live performance and the dancers’ gestures continuously informed each other. Rather than predefined control schemes, the mappings became a shared language, co-shaped in motion—balancing agency, constraint, and improvisation in the unfolding composition.

4 Performance and Deployment

After constructing the garment-based interactive system, we collaborated with two dancers to co-develop a performance that explored the expressive and interactive potential of the embedded textile sensors. Figure 5 presents selected stills from the final performance, illustrating how bodily gestures, fabric stretch, and partner interactions activate the sensor system and shape the audiovisual output in real time.

5 Reflections and Challenges

Integrating conductive materials directly into garments presented unique technical challenges. The proximity to the dancer’s body caused sweat-induced short circuits and unintended conductivity during extended or high-intensity movement. Moreover, the use of a fully soft circuit system introduced persistent noise artifacts due to fabric deformation and non-intentional gestures. These



Figure 4: Snapshots from the performance illustrating body interaction, movement-triggered sound, and the relationship between the two dancers.

conditions made fine-grained signal interpretation unreliable. To overcome this, we shifted our focus toward identifying broader movement trends and implemented threshold-based filtering to extract meaningful patterns from inherently noisy data.

Beyond these technical challenges, we encountered a deeper tension between control and expressivity that is intrinsic to embodied interaction. Drawing on Merleau-Ponty’s phenomenology of movement [6], we came to view the dancer not as a discrete input node, but as a live, sensing subject whose gestures arise from affective, situational, and bodily contingencies. This ambiguity—though problematic from a signal processing perspective—resonates with what Leman [5] terms expressive indeterminacy, in which motion and music co-evolve in a dynamic feedback loop. Rather than eliminating all “noise,” we began to interpret it as part of a corporeal syntax—one that allows gestures to “speak” not through precision, but through visceral intensity. In this way, Dan-sco reframes its goal: from musical control to expressive co-creation, positioning dance not merely as performance, but as a form of embodied sonic authorship.

6 Conclusion and Future Work

Dan-Sco demonstrates the potential of wearable e-textile interfaces to enable dancers to express emotion and generate music through bodily movement. By embedding soft sensors into garments, the system offers a new paradigm for interactive performance that bridges physical expression and real-time sound synthesis.

In future work, we aim to develop a controllable environment interface for training timing-sensitive algorithms that can more accurately recognize dancers’ movement intentions and generate corresponding musical output. We also plan to explore a broader range of movement patterns as expressive modalities, positioning dance not only as performance but also as a framework for emotional communication.

In particular, we hope to extend this project through deeper collaboration with dancers and musicians in co-creative improvisational sessions. These sessions will allow us to explore more diverse mappings between bodily movement and sonic expression, and to uncover new choreographic and compositional possibilities grounded in embodied interaction.

Beyond staged performances, we envision integrating such expressive wearable systems into everyday clothing—for applications in arts education, personal emotional expression, and embodied interaction in dance-based games.

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